

Dehydration of Ethanol to Ethylene

Minhua Zhang and Yingzhe Yu*

Key Laboratory for Green Chemical Technology of the Ministry of Education, Tianjin University Research and Development Center for Petrochemical Technology, Tianjin 300072, People's Republic of China

S Supporting Information

ABSTRACT: This article is an up-to-date review of the literature available on the subject of ethanol to ethylene. The process of ethanol to ethylene has broad development prospects. Compared with the process of petroleum to ethylene, ethanol dehydration to ethylene is economically feasible. Researchers have been redirecting their interest to the ethylene production process, catalysts, and reaction mechanisms. A fluidized bed reactor, together with a wear-resistant, efficient, and stable catalyst will be the focus of future research that includes a deep understanding of the large-scale activated alumina catalyst and the molecular sieve catalyst used, and will promote the development of the ethanol dehydration to ethylene process and provide strong support for the market competitiveness of the process.

1. INTRODUCTION

Ethylene is one of the largest chemical products in the world, and also one of the most important raw materials in the petrochemical industry. At present, about 75% of petrochemical products are produced from ethylene, including acetaldehyde, acetic acid, ethylene oxide, ethylene glycol, ethylbenzene, chloroethanol, vinyl chloride, styrene, ethylene dichloride, and vinyl acetate, etc.; it can also be used as a polymerization raw material to produce a variety of important organic chemical products such as polyethylene, polyvinyl chloride, polystyrene, and so on.¹ Ethylene production has been considered as one of the indicators to measure the petrochemical development level of countries all over the world.

The main technical process to produce ethylene at present is hydrocarbon-cracking. About 99% of the global ethylene is synthesized by this method.² Hydrocarbon-cracking uses petrohydrocarbon or natural gas as raw material. Hydrocarbon compounds with a large number of carbon atoms can be cracked at high temperature into smaller hydrocarbon compounds.³

In recent years, with industrial production capacity increasing year by year, the demand of fossil fuels also has increased in the daily life and industrial production. The nonrenewable fossil fuels will inevitably face the problem of its reserves becoming less and less, and its final depletion is inevitable. Therefore, it is urgent to look for renewable alternative energy sources. Some efforts are being made to research and to develop new methods for using nonpetroleum resources as raw materials to produce ethylene, which is the alternative to traditional hydrocarbon cracking to ethylene process.

According to a survey of the U.S. Department of Agriculture and the European Biomass Industry Association, biomass reserves are huge: the United States, Europe, Africa and Latin America are able to produce the energy of 3.8×10^{10} GJ, 8.3×10^9 GJ, 2.1×10^{10} GJ, and 1.9×10^{10} GJ, respectively. The potential of biomass energy production worldwide in 2050 is expected to be 1.5×10^{11} to 4.5×10^{11} GJ. In the situation of fossil resources becoming increasingly sparse, the use of biomass ethanol catalytic dehydration to ethylene has greater development potential and broad application prospects.^{4–6} Such methods have aroused the attention of the world,

and biomass is considered as one of the most important renewable energy resources.

Ethanol to ethylene reaction occurs through ethanol dehydration under the condition of appropriate temperature and the effect of the catalyst. Ethanol catalytic dehydration to ethylene is the earliest used process in the industry.⁷ X. D. Xu et al.^{8–10} studied the acid-catalyzed dehydration of ethanol, *n*-propanol, ethylene glycol, and glycerol in supercritical water, and found that these dehydration reactions proceed rapidly with a high degree of specificity, making them good candidates for industrial exploitation. The production process of ethanol catalytic dehydration to ethylene is starting to attract more attention. This paper aims to help researchers comprehensively understand the development process and the current status of traditional ethanol catalytic dehydration to ethylene process, including research work on reaction mechanisms and efficient catalyst. The critical problems to be solved by the technology are made clear. The process of petroleum to ethylene and the process of ethanol dehydration to ethylene are preliminarily analyzed and compared on the economic level.

2. ETHYLENE PRODUCTION PROCESS

The American Halcon/Scientific Design Company, base-group of ABB Lummus Global of America and the Brazilian National Petrobras, etc. have developed their own ethanol dehydration technologies. Considering the establishment of the industrial production devices,¹¹ the process of ethanol dehydration to ethylene typically includes two parts: the ethanol dehydration reaction and the purification of ethylene products. Feedstock for ethanol dehydration after preheat evaporation flows into the reactor to generate the crude ethylene, and then into a water washing tower, alkaline washing tower, dryer, light-ends tower, and heavy-ends tower to remove light byproducts and heavy byproducts, etc. Ethylene product is obtained from the overhead

Received: January 21, 2013

Revised: June 18, 2013

Accepted: June 20, 2013

Published: June 20, 2013

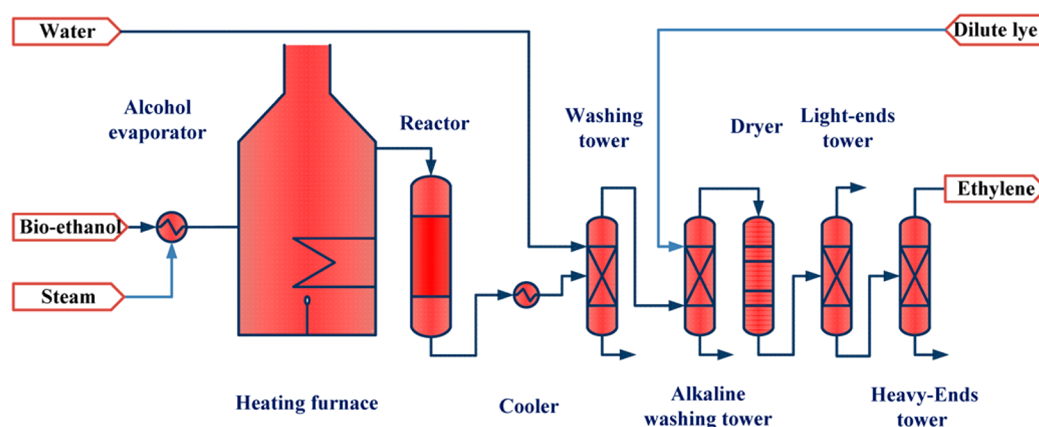


Figure 1. A simple process diagram of a bioethanol-based ethylene plant¹².

of the heavy-ends tower at last. The corresponding process diagram is shown in Figure 1.¹²

Alumina-based catalysts are used in most of the current industrial reactors of ethylene production. Ethanol at the concentration of 95% (w/w) is used as raw material and the operation conditions are a temperature of 300–500 °C, a pressure of 0.1–0.2 MPa, and space velocity of 0.1–1 h⁻¹. The ethylene yield can reach 94–99%.^{13,14} Pearson et al.¹⁵ have made systematic researches on the influencing factors of the process operation of ethanol dehydration to ethylene, including reaction temperature, operating pressure, space velocity, the water content in raw material of ethanol, and so on.

The production process of ethanol dehydration to ethylene has become more mature. The ongoing research work at present has focused on the process intensification, including the improvement of the reactor, process simulation, and optimization.

2.1. Reactor. The reaction of ethanol dehydration to ethylene is an endothermic reaction and it requires relatively more heat, with higher reaction temperature; meanwhile, reaction temperature plays a vital role in the selectivity of the target product, ethylene. The main byproduct generated is ether when the temperature is below 573 K, and the main product ethylene is only generated when the temperature is over 573 K. Therefore, in the process of ethanol dehydration to ethylene, the selection and design of the reactor is critical.¹⁶ There are mainly two kinds of reactors for ethanol dehydration to ethylene, namely, the fixed bed reactor and the fluidized bed reactor. Currently, the fixed bed reactor is mainly used in industrial productions.¹⁷

The fixed bed reactor is the general reactor of a gas–solid catalytic reaction and is widely used in various fields of chemical industry.¹⁸ The fixed bed reactors of ethanol dehydration to ethylene can be divided into two types: isothermal tubular reactor and adiabatic reactor. The isothermal column tube fixed bed reactor was first used in the process of ethanol dehydration to ethylene, and the reactor is most applied in the existing production equipment of ethanol dehydration to ethylene in China. The tubular fixed bed reactor has relatively poor temperature control ability, and regeneration and replacement of the catalyst is also more complex. In addition, the reactor is very sensitive to small changes of the reaction process. For example, slight changes of raw material temperature are likely to lead reactors to be out of control. Researchers call this phenomenon as “parametric sensitivity”.¹⁹ From 1950s to 1980s, many researchers conducted the study on tube fixed bed reactor, but they did not thoroughly solve the problems existing in the reactor.^{20–22} Froment et al.²³ consider that the adiabatic fixed bed reactor is the most basic

form of fixed bed reactor. When it is used for the endothermic reaction, the raw material should enter the reactor together with the heat carrier and the reaction should occur step by step; that is, a multiple adiabatic fixed bed reactor should be used to heat the reactants between the two reactors, which can effectively ensure that the reactors reach the reaction temperature so that the reaction can be carried out smoothly. Barrocas et al.²⁴ adopted multiparallel or a parallel series of adiabatic fixed bed reactor for the reaction of ethanol catalytic dehydration to ethylene at high temperature, and the heat required by the reaction was provided by the steam entering the reactor together with raw materials. The experimental results show that the reactor effectively improved the ethanol conversion rate and ethylene yield.

The fluidized bed reactor has a relatively short development history. It can be used in a multiphase chemical reaction, and it has such advantages as high heat and mass transfer rate, more uniform bed temperature, and relatively stable operation. It is particularly suitable for large-scale production with great thermal effect, and it has been applied to the production equipment of many chemical products.^{25,26} Tsao et al.²⁷ have developed a reaction process of ethanol catalytic dehydration to ethylene in the fluidized bed reactor using a Si–Al-oxide catalyst. The one-way ethanol conversion rate was higher than 99.5%, and the ethylene yield was greater than 99% before distillation, values which were consistently higher than that of the tube fixed bed process. The process of the fluidized bed reactor of ethanol dehydration to ethylene has a promising development prospect, but there are also urgent problems to be solved; that is, the friction and collision between catalyst particles and those between catalyst particles and the reactor walls are likely to wear the Si–Al-oxide catalyst, and wear-resistance of the catalyst should be improved.

2.2. Simulation and Optimization of Process. The chemical process simulation is the basic technology in process systems engineering. Process analysis and optimization is undertaken on the basis of process simulation.²⁸

The process simulation method was used to simulate the process of ethanol dehydration to ethylene, which can improve the heat recovery capacity of the system, reduce the auxiliary heating and cooling load (such as: steam, cooling water, etc.) of the utility as much as possible, improve the energy efficiency and economy of the overall process, reduce the production cost and further improve the competitiveness of biomass–ethanol dehydration to the ethylene process. Wang J.²⁹ adopted the Aspen Tech process simulation software to conduct simulation of design conditions and transformation conditions for the

distillation section of the ethanol and ethanol dehydration to ethylene in a domestic plant. Energy-saving improvement was conducted on the basis of the existing process. The ethanol concentration of the rectifying section in the ethanol production process was reduced to 46.4%. Steam did not need the addition of a carrier gas in the dehydration section of the improved process, the energy consumption of which was 20.2% less than that of the existing process. Chen Yu¹² has used the PRO/II process simulation software and conducted a simulation of ethanol to ethylene process in a domestic plant. The researchers used double-effect distillation technology to improve the existing process, and adopted the technology of a heat exchanging network on this basis to optimize processes and reduce energy consumption. After optimization, the heating utility was reduced by 8.23%, cooling utility by 5.85%, and compressor work by 24.13% per unit compared to existing process.

Aslo, researchers used the process simulation method to conduct auxiliary study on the operation conditions of the ethanol dehydration to ethylene reactor. Kagymanova et al.³⁰ have researched the reaction of the ethanol dehydration to ethylene on the commercialized Al-based catalyst from multiple aspects, including a reaction kinetics study, the reactor testing, and process simulation. On the basis of the process simulation, operating characteristics and optimal operating conditions of the ethanol dehydration industrial tubular fixed bed reactor with an annual output of 60 000 tons of ethylene have been determined.

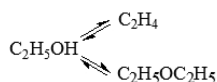
3. REACTION MECHANISMS

Since the middle of 20th century, many researchers have conducted studies on the reaction mechanism of ethanol dehydration with different catalysts, including activated alumina,^{31,32} phosphate,³³ magnesium oxide,³⁴ molecular sieve,^{35,36} heteropoly acid,^{37,38} and so on, but up to now, there has still been no consensus, and the research in this field is continuing.

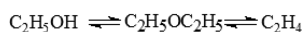
In addition to the main product of ethylene and the main byproduct of ether, the reaction of ethanol dehydration may also generate a small amount of byproducts, such as acetaldehyde,³⁹ hydrocarbons (methane, ethane, propylene, butylene),⁴⁰ and light base-groups⁴¹ (CO₂, CO, H₂, etc.) and so on. As the amount of other byproducts is small, most of the mechanism research of ethanol dehydration reaction considers mainly the generation of ethylene and ether, which can be summarized as three kinds of routes: (1) parallel reactions,⁴² (2) a series of reactions, and (3) a parallel series reaction. The main controversy lies in whether ethylene is directly generated from ethanol or indirectly generated from ether, or both routes coexist, just as is shown in the following three formulas.

(1) Parallel reactions

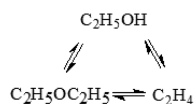
1) Parallel reactions



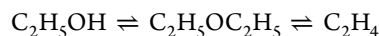
2) A series of reactions



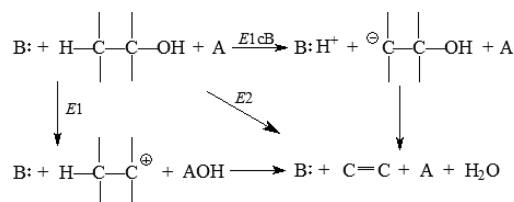
3) A parallel series reactions



(2) A series of reactions



(3) A parallel series reaction

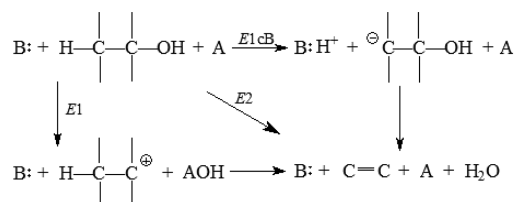


: electrons pair $\oplus$ positive charge $\ominus$ negative charge

The above three reaction routes refer to three reversible reactions, namely ethanol intramolecule dehydration to ethylene, ethanol intermolecule dehydration to ether, and ether dehydration to ethylene.

3.1. Ethanol Intramolecule Dehydration to Ethylene.

There are mainly three kinds of reaction mechanisms of ethanol catalytic dehydration to olefin under different reaction conditions,⁴³ namely, E1, E2, and E1cB mechanism, as is shown as follows, where A and B are the acidic and basic centers of the catalyst, respectively.



: electrons pair $\oplus$ positive charge $\ominus$ negative charge

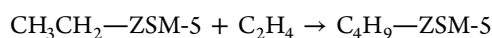
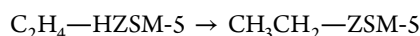
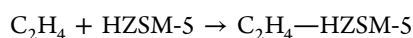
The reactions of E1, E2, and E1cB are elimination reactions,⁴⁴ and the three are competitive reactions. The E1 reaction is a single-molecule elimination reaction, and it first generates a carbonation intermediate, which is the rate controlling step and is a first-order reaction, then quickly loses β -hydrogen to generate olefin. The E2 reaction is a bimolecular elimination reaction, and the reaction is finished in one step; the reaction rate is influenced by the concentration of the two compounds, which is a second-order reaction. E1cB reaction is a single-molecule conjugate base elimination reaction; in the reaction process, the nucleophilic center first captures β -hydrogen of the reactant to generate carbanion (conjugate base), and then the hydroxyl of the conjugate base leaves to generate olefin, which is the first step reaction and should be an equilibrium reaction with a quicker rate. The second step is the rate-limiting step in the overall reaction with a slower rate, influenced only by the concentration of one kind of molecule.

Noller et al.⁴⁵ have studied the ethanol dehydration process. They found that the carbocations in the E1 reaction will not generate ethanol, the E2 reaction is not reversible, and the carbanion of the E1cB reaction can generate ethanol. The reaction mechanism of ethanol dehydration process is influenced by such factors as ethanol, catalyst type, and so on.

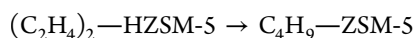
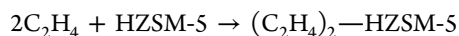
The active sites of ethanol intramolecular dehydration to ethylene are the weak acid centers and relatively strong acid centers of the catalyst. However, the strong acid centers can easily lead to ethylene polymerization, which is of great harm to the ethanol dehydration reaction, especially the stability of the reaction. Ramesh et al.⁴⁶ prepared modified ZSM-5 zeolites with different P content and used them in ethanol dehydration to

ethylene reaction. The results of the research suggest that the introduction of P will block the pores, reduce the adsorption of the polymer, and thus improve the selectivity and thermostability. In ZSM-5 with P introduced, no ethylene polymer and aromatic hydrocarbons are found and coke resistance performance of the catalyst is improved. Zhang et al.⁴⁷ conducted a theoretical study on the reaction mechanism of ethylene dimerization on ZSM-5 zeolite and they argued that there are two main reaction mechanisms of ethylene dimerization on HZSM-5 zeolite: step-by-step reaction route and concerted reaction route.

a. Step-by-step Reaction. In the first step the ethylene molecule adsorbs on the Brønsted acid sites of HZSM-5 zeolite via hydrogen bonds. In the second step ethylene molecules generate an ethoxide intermediate via protonation. In the third step the second ethylene molecule adsorbs on the zeolite catalyst, attracts the ethoxide intermediate, then generates a C–C bond and a new C–O covalent bond with the ethoxide intermediate, and finally generates butoxide. The reaction steps are shown as follows:



b. Concerted Reaction. First of all, two ethylene molecules coadsorb on the acid site of zeolite catalyst. Then, ethylene is protonized; the generation of C–C bond and C–O bond occurs simultaneously, generating butoxide. The reaction steps are shown as follows:



3.2. Ethanol Intermolecular Dehydration to Ether.

Knozinger^{48,49} has summarized the research progress of the ethanol dehydration reaction on $\gamma\text{-Al}_2\text{O}_3$. When the temperature was lower than 513 K, the ethanol dehydration mainly generated ethyl ether; when the temperature was 513–523 K, the ethyl ether would be converted to ethanol; when the temperature was above 513 K, the ethanol dehydration generated the main product of ethylene. During the process, when ethanol dehydration reaction generates ether, the infrared spectroscopy on the catalyst surface only detects the intermediate product alkoxide. Alkoxides (hydroxyl hydrogen of the ethanol molecule is substituted by the metal compounds) is the necessary condition for the generation of ether, suggesting that ether has two possible formation mechanisms: one is that the hydroxyl of R–OH adsorbed on the catalyst surface leaves to react with RO–Al– adsorbed on the Al to generate ether. The other is that the R–O bond of the R–OH of the alcohol molecule and R–O bond of the R–O–Al– adsorbed on Al break, and both generate ether.

Arai et al.⁵⁰ have studied the ethanol dehydration reaction on the Si–Al composite oxide catalyst, and they think that two RO–Al– adsorbed on the catalyst surface Al atoms react; that is, the O–Al bond breaks and the R–O bond also breaks, and then both react to generate ether.

Jain et al.⁵¹ have studied the reaction of alcohol dehydration on alumina, and found that there are acid centers and also alkaline centers on the alumina surface; the C–O bond of the alcohol molecules adsorbed on the acid sites breaks, and the O–H bond

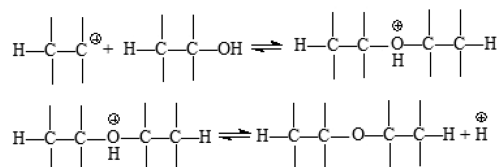
in the alcohol molecules adsorbed on base sites breaks, and then both react to generate ether.

El-Salaam et al.⁵² have researched the reaction of the ethanol dehydration on CdO, and found that although alcohol dehydration is a catalytic reaction of the acid centers, the existence of a pH center at the same time is more conducive to the reaction. Ethanol molecules adsorbed on acid sites lose hydroxyl, and hydroxyl of ethanol molecules adsorbed on base sites loses hydrogen, and then both react to generate the ether.

From the above studies, we can find that the reaction mechanism of the ethanol to ether has not reached a consensus, but it can be ensured that the alkoxide is an important intermediate of the ethanol to ether; the reaction of the ethanol intermolecular dehydration to ether is a substitution reaction. The generation of ether essentially follows the reaction mechanism of $\text{S}_{\text{N}}1$ (single-molecule nucleophilic substitution reaction) or $\text{S}_{\text{N}}2$ (bimolecular nucleophilic substitution reaction).⁴⁴

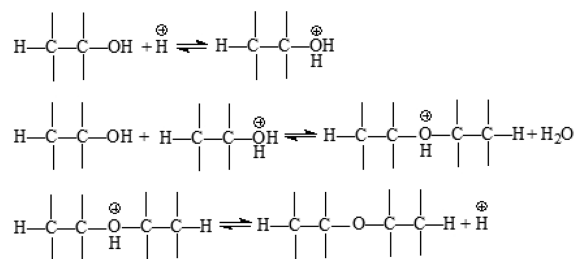
The $\text{S}_{\text{N}}1$ reaction is divided into two steps: the first step is that reactants dissociate to carbocations and the negatively charged leaving group, which is the rate-controlling step and is the same as the first reaction step of mechanism E1; the second step is the carbocations associate with nucleophiles with an extremely fast rate, which is a first-order reaction. In the $\text{S}_{\text{N}}2$ reaction, lone pair electrons of the nucleophile attack the electrophilic electron-deficient central atom, forming the intermediate, and losing the leaving base-group at the same time; there's no carbocations generated in the reaction, and the rate controlling step is the joint of both steps, and it is a second-order reaction. The $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ reaction mechanisms of ethanol intermolecular dehydration to ether are shown as follows.

1) $\text{S}_{\text{N}}1$ reaction mechanism (the first step is the same as that of E1)



⊕ positive charge

2) $\text{S}_{\text{N}}2$ reaction mechanism



⊕ positive charge

The oxygen in ether molecule has an unshared electron pair, which can be considered as a weak alkali. When the ether adsorbs on the acid sites of the catalyst, oxygen onium salt is formed, and then the carbon–oxygen bond becomes weak. At this time, with the temperature increasing, the ether bond will break.⁵³ The ether bond breaks to generate a carbocation and may further generate ethylene, following the E1 reaction mechanism.

4. CATALYSTS

The ethanol dehydration to ethylene is an acid-catalyzed reaction. Such catalysts have been researched and developed as clay, activated alumina, silicon oxide, magnesium oxide, zirconium oxide, phosphate, calcium phosphate, zinc aluminates, Heteropoly salt, and molecular sieve, etc., which can be summarized as four categories: phosphoric acid, oxides, molecular sieves, and miscellaneous acid catalyst.⁵⁴

4.1. Catalyst of Phosphoric Acid System. The first catalyst used in the industrial equipment of ethanol catalytic dehydration to ethylene plant was the catalyst of phosphoric acids. In the 1930s, it was developed by the British Imperial Chemical (ICI) by loading phosphate on clay or coke.⁵⁵ Donald et al.⁵⁶ have researched the reaction temperature, alcohol concentration, space velocity, and the impact of the catalyst regeneration times on the reaction of ethanol catalytic dehydration to ethylene with phosphoric acid catalyst. The ethylene from ethanol dehydration reaction with phosphoric acid catalyst is of high purity; however, the catalyst is easily deactivated by coke deposition, and it requires about a month to regenerate. Since the 1950s, such catalysts have no longer been used.

4.2. Oxide Catalyst. The oxide catalyst of ethanol dehydration is the typical representative of activated alumina-based catalysts. The catalyst currently used in the industrial equipment of the ethanol dehydration to ethylene is mainly the activated alumina-based catalyst.⁵⁷ Alumina is an important catalyst or catalyst carrier. It has long been found that alumina could be used for the alcohol dehydration reaction and also used in many other chemical reactions such as isomerization, alkylation, and catalytic cracking, etc.⁵⁸

In the 1950s, there was a lot of production equipment for ethanol dehydration to ethylene built in Asia and South America, mainly using clay or alumina as the catalyst. Comparison of the two catalysts shows that the deactivation of clay is more serious than that of alumina, while the ethylene product obtained from alumina catalyzing ethanol dehydration is of relatively low purity, which is about 96% to 97%.⁵⁹ In 1960, when Lummus in America built the industrial plant of ethanol to ethylene in India, γ - Al_2O_3 was used as the catalyst, and the ethanol conversion rate was up to 99%, and the yield of ethylene was 94%.⁶⁰ Roca et al.⁶¹ have investigated the ethanol dehydration reaction on the Si–Al catalyst, and they found that ethylene was the main product, and ethylene and the byproduct ether followed the parallel reaction mechanism. In the 1980s, American Halcon Scientific Design Inc. successfully developed a new type of multioxide catalysts with the main components of Al_2O_3 – MgO /SiO₂, which were named the Syndol catalyst and were used in industrial equivalent temperature fixed bed reactors; the ethanol conversion rate was 97%–99%, and the product ethylene selectivity was nearly 97%; the one-way cycle of the catalyst was 8–12 months.⁶² The Syndol catalyst is considered as a kind of highly effective catalyst for ethanol dehydration to ethylene. Kochar et al.⁶³ have investigated the reaction of ethanol dehydration to ethylene with Syndol catalyst, and found that the rising of the reaction temperature was conducive to the improvement of ethylene selectivity. Ezzo et al.^{64–66} performed further research on the Al_2O_3 – Cr_2O_3 catalyst, put the catalyst in an ethanol catalytic conversion reaction, and found that the main gas product was ethylene and the liquid products were mainly water, ether, and a small amount of acetaldehyde and acetone. The addition of Cr_2O_3 improved the activity of the catalyst for ethanol dehydration, and the generation rate of ethylene on the catalyst surface was fastest at 673 K. In the study, Kojima et al.⁶⁷ found that the catalyst for the

ethanol dehydration to ethylene with a higher conversion rate of raw materials and the selectivity of target products can be prepared by adding a small amount of Ca, Mg, or Zn phosphate in the alumina. El-Katatny et al.⁶⁸ have researched the reaction of ethanol dehydration to ethylene on $\text{FeOx}/\text{Al}_2\text{O}_3$ catalyst, and the results showed that the catalyst did not change the conversion rate of raw materials of ethanol, but it improved the product ethylene selectivity. Chen et al.⁶⁹ adopted $\text{TiO}_2/\gamma\text{-Al}_2\text{O}_3$ to catalyze the bioethanol dehydration to an ethylene reaction in a microchannel reactor with the ethanol conversion rate of 99.96% and the ethylene selectivity of 99.4%, which showed that the addition of TiO_2 helped to improve the ethanol conversion rate and ethylene selectivity. Y. X. Yu et al.⁷⁰ researched and developed the NC1301 spherical catalyst with alumina as the main component, and put the catalyst into practice in industrial production equipment in the Changzhou Chemical Factory. The results showed that when the reaction temperature was between 623 and 713 K, the pressure was lower or equal than 0.3 MPa and the space velocity was between 0.3 and 0.6 h^{−1}; the ethanol conversion rate was 99.53–100% and the ethylene selectivity was 99.57–100%, with a life cycle of 12–18 months and 1–2 times of regeneration during that period. Y. Li et al.⁷¹ investigated the activated-alumina catalyzing low concentration ethanol dehydration reaction in a tubular fixed bed reactor. When the volume concentration of the raw material of the ethanol changes from 50% to 100%, the ethanol conversion rate increased from 88% to 98%, and the selectivity of target product ethylene was over 98% on average.

The activated alumina-based catalyst has good stability, and the purity of produced ethylene product based on such a catalyst is also higher. But the concentration of raw material of ethanol should not be too low, or it can make the ethanol dehydration reaction require higher temperature and lower space velocity, which in turn leads to higher energy consumption.

4.3. Molecular Sieve Catalysts. Molecular sieve has the regular pore structure, unique acid–base properties, and large specific surface area, which can be used as adsorbent material, catalytic material, and ion exchange material, and in many other fields, such as petrochemical, environmental decontamination, and detergent industry, etc.⁷² Since the 1980s, researchers have applied the molecular sieves for the ethanol catalytic dehydration to ethylene reaction, and the molecular sieves mainly used include ZSM-5 type, the Si–Al-phosphate (SAPO) type, A type, and AM-11 type, etc., of which, the most studied one is the ZSM-5 molecular sieve catalyst. SAPO series molecular sieves have good catalytic ability for methanol dehydration to light olefins, and they also have attracted the attention of more researchers.⁷³

4.3.1. ZSM-5 Molecular Sieve Catalysts. In the 1970s, Argauer et al.⁷⁴ and Dwyer et al.⁷⁵ successfully synthesized ZSM-5 molecular sieve. The ZSM-5 zeolite molecular sieve has a two-dimensional 10-membered ring structure (0.53–0.56 nm × 0.51–0.55 nm), excellent shape-selective catalysis, and is one of the most important molecular sieve catalytic materials, widely used in petrochemical, coal chemical industry, fine chemical catalysis, and other catalytic fields.^{76,77}

At the first stage of these researches, the researchers used pure ZSM-5 zeolite as the catalyst for ethanol dehydration to ethylene reaction. Hao Tong⁷⁸ studied the reaction of a dilute ethanol solution (20% v/v) to ethylene on ZSM-5 zeolite, and found that when the reaction temperature was 672 K, the ethanol conversion rate was about 99%, with the ethylene selectivity of about 80%; however, when the reaction temperature was 571 K, the ethanol conversion rate decreased to about 42%, with an ethylene

selectivity of about 72%. Le Van Mao et al.⁷⁹ put forward the method of bioethanol fermentation with the HZSM-5 zeolite catalyst (without fine distillation) to generate ethylene in one step, which was called B.E.T.E. (bioethanol-to-ethylene). The feed ethanol concentration was 15%, and when the reaction temperature reached 673 K, the ethanol conversion rate was 96%, and ethylene selectivity was 49%. Phillips et al.⁸⁰ studied the dehydration reaction on HZSM-5 under mild conditions of pressure below 0.07 MPa and temperature less than 473 K, by using an ethanol aqueous solution of different volume concentration (80–100%) as the raw material, and found that carbon deposition is more likely on the HZSM-5 catalyst, and water in ethanol helps to improve the catalyst activity and inhibit carbon deposition. The research results at this stage showed that ZSM-5 zeolite could be used to catalyze the ethanol dehydration reaction. However, the reaction should be carried out at higher temperature, and the catalyst gets deactivated easily. To solve the aforementioned problems, the researchers modified the ZSM-5 zeolite with the hot water treatment, ion exchange, impregnation, and other methods. Pan, L. V. et al.^{81,82} modified HZSM-5 zeolite and then developed NKC-03A catalyst, using industrial ethanol as the raw material, when the temperature was 533 K, space velocity was 2.3 h⁻¹, the ethanol conversion rate was 98%, and ethylene selectivity was 98–99%. The catalyst has been used in industrialized column tube reactors. Le Van Mao et al.⁸³ studied the ethanol catalytic dehydration reaction with ZSM-5 zeolite modified with zinc and manganese, and found that when the temperature was 673 K with space velocity of 2.5 h⁻¹, the liquid ethanol content of raw material was 75%, the ethanol conversion rate was 99% and C2~C4 olefin yield was 94%, which mainly was ethylene. Bun et al.⁸⁴ used ZSM-5 zeolite modified with H, Cu ion for ethanol conversion reaction separately, and the results showed that the H-ZSM-5 catalyzed ethanol dehydrogenation effectively, while the ethanol on Cu-ZSM-5 zeolite was deeply oxidized to generate CO₂ and CO; increasing the hydroxyl on the H-NaZSM-5 surface can enhance the dehydrogenation activity and decrease oxidation activity. Increasing Cu ions on the Cu-NaZSM-5 surface is good for the ethanol catalytic oxidation reaction. Takahara et al.⁸⁵ researched the reaction of the ethanol dehydration to ethylene on different solid acid catalysts, including mordenite, ZSM-5 zeolite, H β hydrogen type zeolite, and Si–Al compounds, and the results showed that the catalyst activity was related to the number of B acid sites, and mordenite had the highest catalytic activity for alcohol dehydrogenation. Jia, O. Y. et al.⁸⁶ conducted the ethanol dehydration reaction on ZSM-5 zeolite catalyst modified with La in the laboratory bioreactor. During the reaction over 950 h, the ethanol conversion rate and ethylene selectivity were above 98%, and the time required for the catalyst regeneration was no more than 830 h. Ramesh et al.⁸⁷ used the impregnation method to prepare the HZSM-5 catalyst modified with P, the evaluation test of ethanol catalytic dehydration was done at the temperature ranging from 523 to 723 K. The catalytic activity was influenced by P content, reaction temperature, and space velocity, with no deactivation within 200 h; at a temperature of 623 K, the ethanol solution with a concentration of 10% was used as the raw material, and ethylene selectivity was above 98%; Characterization results of the catalyst showed that the addition of P decreased ZSM-5 acidity.

4.3.2. SAPO Series of Molecular Sieve. Union Carbide (UCC)⁸⁸ in America developed a series of molecular sieve of Si–Al-phosphate (SAPO-*n*, *n* represents the model) in 1984. The composition of the series of molecular sieves can change in a wide range and the silicon content changes with the synthesis conditions. In the SAPO series of molecular sieve, SAPO-34 molecular sieves have eight-member pore structure of chabazite,

and the effective orifice diameter is between 0.43 and 0.50 nm, showing excellent catalytic performance in the methanol-to-olefin (MTO) reactions.⁸⁹ D. Y. Wang et al.⁹⁰ researched the reaction of ethanol dehydration in a SAPO-34 molecular sieve under the conditions of space velocity of 2 h⁻¹ and temperatures of 493–593 K, and the results showed that when the reaction temperature of SAPO-34 catalyst was higher than 533 K, the ethanol conversion rate was about 90% and ethylene selectivity was about 99%; and the ether was generated when the reaction temperature was below 533 K or space velocity was larger. Under the reaction conditions (reaction condition: temperature of 49–533 K, LHSV 2 h⁻¹), the catalytic activity of SAPO-34 was better than that of ZSM-5. Under the conditions of temperature of 673 K, mass ratio of ethanol/water of 1, the pressure of ethanol and moisture of 0.4 MPa (carrier gas is Ar), and space velocity of 16.7 h⁻¹, Dahl et al.⁹¹ studied the reaction of ethanol dehydration on SAPO-34 and found that ethanol was quickly translated into ethylene and water on SAPO-34 pretreated by propylene. The ethylene conversion process is not diffusion-limited under these reaction conditions. Zhang et al.⁹² compared the activity and stability of four kinds of catalysts in the reaction of ethanol dehydration to ethylene, namely γ -Al₂O₃, HZSM-5 (Si/Al = 25), SAPO-34, and NiAPSO-34, and found that the activity of the catalysts in descending order was HZSM-5 > NiAPSO-34 > SAPO-34 > γ -Al₂O₃ (reaction condition: temperature 723 K (γ -Al₂O₃), 573 K (HZSM-5), and 623 K (SAPO-34 and NiAPSO-34), LHSV 3 h⁻¹), while in the stability test under 100 h, SAPO-34 and NiAPSO-34 have the best stability. After considering all the aspects, NiAPSO-34 is most suitable for the ethanol catalytic dehydration to ethylene reaction. Zhou T. et al.⁹³ studied the ethanol (mass concentration is 99.7%) dehydration to ethylene reaction on the composite catalyst of SAPO-11/HZSM-5 and found that the increase of the catalyst acidity helped to improve the ethanol conversion rate and ethylene yield, when the pressure was 0.1 MPa, the temperature was 513 K, the space velocity was 1.2 h⁻¹, the ethanol conversion rate was 99.19%, and ethylene selectivity was 98.77%; when the ethanol conversion rate and ethylene selectivity both reached 99.00%, the reaction temperature of the catalyst was 60 K lower than those of HZSM-5 and SAPO-11.

The activity of ethanol catalytic dehydration reaction of various types of molecular sieve is not very high, which must modulate the surface acidity and pore size to improve the catalytic activity through modified molecular sieve.⁹⁴ Compared with the activity of activated alumina-based catalysts, the modified molecular sieve catalyst has higher activity, its reaction temperature is lower, and it can catalyze the ethanol aqueous solution of low concentration for the dehydration reaction, but it is less stable, and it can be deactivated easily with high cost and complicated preparation process, which is still far from applicable for large-scale industrial production.

4.4. Heteropolyacid Catalyst. Heteropolyacid is the general term of an oxygen-containing multiacid formed by the central atom (such as P, Si, Ge, Fe, and Co) and the lig-atom (such as Mo, W, V, and Nb, etc.) through the oxygen atom bridging. These catalysts can be used as the acid or the oxidation–reduction catalyst and can be used both in the homogeneous reaction or nonhomogeneous reaction.^{95,96}

Zhao B. L. et al.⁹⁷ have researched the reaction of ethanol dehydration to ethylene on six different heteropoly acids, and found that when the temperature was over 443 K, ethylene was the main product, and when the temperature was below 443 K, ether is the main product; in the 440 h' evaluation test, the catalyst was stable without deactivation. Vázquez et al.⁹⁸ have made Keggin-type phosphor-molybdate and phosphor-tungstic

acid loaded on SiO_2 , and used them in a liquid dehydration reaction of 96 wt % ethanol solution, and they carried out the full reaction at 293 K for 72 h, and they found that the catalytic activity is satisfactory, and the catalyst can be separated from the system through the centrifugal separation method; the catalyst used for several times still had good catalytic activity, and the characterization results showed that there's basically no structure difference between the used catalyst and the original catalyst. Haber et al.⁹⁹ have added K and Ag to phosphotungstic acid (HPW) and made two phosphotung salt of $\text{K}_x\text{H}_{3-x}\text{PW}_{12}\text{O}_{40}$ and $\text{Ag}_x\text{H}_{3-x}\text{PW}_{12}\text{O}_{40}$, and loaded them on SiO_2 under the conditions of atmospheric pressure and temperature of 398–773 K for the ethanol dehydration reaction on it. When the temperature is lower than 523 K, the main product on $\text{Ag}_x\text{H}_{3-x}\text{PW}_{12}\text{O}_{40}/\text{SiO}_2$ is ethylene, and the yield increases up to about 45% with temperature rising; when the temperature is between 523 and 573 K, the main product is acetaldehyde; when the temperatures continues to rise, the ethylene yield rises gradually, after it reaches over 673 K, the ethylene yield will increase faster up to about 70% with temperature rising. The activity of $\text{K}_x\text{H}_{3-x}\text{PW}_{12}\text{O}_{40}/\text{SiO}_2$ is low, close to the dehydration activity of SiO_2 . Varisli et al.¹⁰⁰ have investigated the ethanol dehydration to ethylene reaction and ethanol to ether reaction (ethanol was 5 wt %, and the remaining was He gas) on three kinds of heteropolyacid catalysts of phosphor-tungstic acid (TPA), silico-tungstic acid (STA), and molybdo-phosphoric acid (MPA) at the temperature ranging from 413 to 523 K, and the results showed that the water vapor would reduce the activity of the catalyst; it generated ether when the temperature was below 453 K; ethylene yield on heteropolyacid catalyst can reach 77%; the activity of three heteropoly acid catalysts in descending order was $\text{STA} > \text{TPA} > \text{MPA}$.

Heteropolyacid catalyst used for ethanol dehydration to ethylene has the advantages of low reaction temperature, etc., but the ethanol conversion rate is relatively low. When used, Heteropolyacid catalysts normally need to be loaded on the carrier, with the problems of serious loss and high preparation cost. Thus, further study on Heteropoly acid catalyst is needed.

5. PRELIMINARY ANALYSIS OF ECONOMY

The economic analysis and discussions on ethanol dehydration to ethylene are mainly based on the production process of the American Halcon/Scientific Design Company.¹⁰¹ The components of ethylene product with different purity are shown in Table S1 in the Supporting Information. The investment (areas beyond the battery are not included), raw materials, labor and utilities consumption for producing ethylene with different purity are shown in Table S2 and Table S3 in the Supporting Information. In the ethylene production process of the American Halcon/Scientific Design Company, the investment of different reactor systems is shown in Table S4 in the Supporting Information. Assuming that ethylene production is 17 300 tons per year, ethanol is based on the wood of 700 yuan per ton, and other raw materials, chemicals and utilities all refer to the consumption quota of intermedia ethylene, the calculated production cost of ethylene is roughly 1900 yuan per ton or so. The plant investment of the two processes of hydrocarbon thermal cracking to ethylene and ethanol dehydration to ethylene were once compared by the base-group of ABB Lummus Global of America, as is shown in Table S5 in the Supporting Information.

From the figures in the table, the investment of a unit of ethylene of ethanol dehydration is lower than that of hydrocarbon thermal cracking. Compared with the ethylene

production process through petroleum, the biomass ethanol dehydration to ethylene process has the following advantages:¹⁰²

- (1) The purity of bioethylene is high; the cost of separating and refining ethylene is very low; the investment is small; the construction period is short; and returns are fast.
- (2) The raw materials are easily available; it is not limited by the resource distribution; and it can promote the development of cyclic economy.
- (3) Complex technologies or equipment are not needed; the process can be easily improved.

In the age of shortage of petroleum resources, biomass ethanol dehydration to ethylene is sure to become a very promising chemical process.

6. CONCLUSIONS

The development level of the ethylene industry is one of the important indicators to measure the economic power of a country, and plays a vital role in the national economic development. That petroleum resources are becoming increasingly strained restricts the development of the ethylene industry. The development and utilization of various nonpetroleum raw materials producing ethylene have drawn extensive attention of the countries. The process of ethanol to ethylene has broad development prospects and has great significance for promoting the economy development. Compared with the process of petroleum to ethylene, ethanol dehydration to ethylene is economically feasible.

At present, the studies of ethanol dehydration to ethylene mainly focus on such aspects as the production process, the reaction mechanism, and highly efficient catalyst.

The reactor is the key equipment of the whole technology process of ethanol dehydration to ethylene. At present, compared with the fixed bed reactor which is most commonly used in the industrial equipment, the fluidized bed reactor has such advantages as larger heat and mass transfer rate, more uniform bed temperature, and more stable operation. It is especially suitable for large-scale production with great thermal effect. However, the catalyst is easily worn due to the friction and collision between the catalyst particles and between the catalyst particles and the reactor. The fluidized bed reactor, together with a wear-resistant, efficient, and stable catalyst will be the focus of future research.

The catalyst is an effective means to change the reaction rate, and control the direction of the reaction. A further understanding of the reaction mechanism can help design and filter the basic component of the catalyst. At present, the catalyst mainly applied in the industrial plant of ethanol dehydration to ethylene is an activated Al_2O_3 -based catalyst. The reaction temperature is relatively high and some of byproducts are unfavorable for the production of the downstream products. Zeolite catalyst is the focus of current research, with which ethanol catalytic dehydration can occur at lower temperature. However, its stability is poor, and it can be easily deactivated through carbon deposition. Its cost is high and its preparation process is complex. Thus, its usage is still at the exploratory stage in the laboratory with some distance from large-scale industrial production application. On one hand, the future study will be focused on deeply understanding the large-scale used activated alumina catalyst with a reasonable proposal for design and modification being put forward to reduce the reaction temperature and to optimize the product distribution. On the other hand, efforts should be made to improve the stability of the zeolite catalyst to

promote its industrial application to promote the development of ethanol dehydration to the ethylene process and provide strong support for the market competitiveness of the process.

■ ASSOCIATED CONTENT

■ Supporting Information

Components of each kind of ethylene product (S1), the plant investment of an annual output of 50 000 tons of ethylene (S2), the requirement of raw materials and utility of equipment of an annual output of 50 000 tons of ethylene (S3), the investment of the reactor of an annual output of 50 000 tons of ethylene (S4), investment comparison of ethylene plants (S5). This material is available free of charge via the Internet at <http://pubs.acs.org>.

■ AUTHOR INFORMATION

Corresponding Author

*Tel.: +86-22-27406119. Fax: +86-22-27406119. E-mail: yuyingzhe@tju.edu.cn

Notes

The authors declare no competing financial interest.

■ REFERENCES

- (1) Huang, L. D. Ethylene and its industry status. *China Chlor-Alkali*. 2005, 5 (5), 1–5.
- (2) Wang, S. H.; He, X. O. *Ethylene Process and Technology*; China Petrochemical Press: Beijing, 2000.
- (3) Okagami, A.; Matsuoka, S. Process for manufacturing olefins by catalytic oxidation of hydrocarbons. US patent 3541179, 1970–11–07.
- (4) Sheehan, J. A. *A Strategic Perspective of the U.S. Department of Energy's National Ethanol Program*; American Chemical Society: Washington, NW, 1997.
- (5) Bastianoni, S.; Marchettini, N. Ethanol production from biomass: Analysis of process efficiency and sustainability. *Biomass Bioenergy* 1996, 11 (5), 411–418.
- (6) George, W. H.; Sara, I.; Avelino, C. Synthesis of transportation fuels from biomass: Chemistry, catalysts, and engineering. *Chem. Rev.* 2006, 106 (9), 4044–4098.
- (7) Olaf, W.; Meng-Teck, E. Make ethylene from ethanol. *Hydrocarbon Process.* 1976, 55 (11), 125–133.
- (8) Xu, X. D.; Almeida, C. D.; Antal, M. J., Jr. Mechanism and kinetics of the acid-catalyzed dehydration of ethanol in supercritical water. *J. Supercrit. Fluids* 1990, 3, 228–232.
- (9) Xu, X. D.; Almeida, C. D.; Antal, M. J., Jr. Mechanism and kinetics of the acid-catalyzed formation of ethene and diethyl ether from ethanol in supercritical water. *Ind. Eng. Chem. Res.* 1991, 30 (7), 1478–1485.
- (10) Ramayya, S.; Brittain, A.; DeAlmeida, C.; Mok, W.; M. J., A., Jr. Acid-catalyzed dehydration of alcohols in supercritical water. *Fuel* 1987, 66, 1364–1371.
- (11) Braskem, S. A. Bio-ethanol based ethylene. *J. Macromol. Sci.* 2009, 49 (2), 79–84.
- (12) Chen, Y. Simulation and Synthesis of Heat Exchange Networks for Producing Ethylene from Ethanol. Tianjin University: Tianjin, China, 2007.
- (13) Zhang, W. G. Studies on catalysts and technology for catalytic dehydration of ethanol to ethylene and aromatics. Hunan University: Changsha, China, 2008.
- (14) Sundaram, K. M.; Shreehan, M. M.; Olszewski, E. F. *Kirk-Othmer Encyclopedia of Chemical Technology: Ethylene*; Wiley and Sons: New York, 2010.
- (15) Pearson, D. E. Process for catalytic dehydration of ethanol vapor to ethylene. US Patent 4423270, 1983-12-27.
- (16) De Moraes, E. R.; Lunelli, B. H.; Jaimes, R. R.; De Souza Victorino, I. R.; Maciel, M. R. W.; Filho, R. M. Development of an industrial multitubular fixed bed catalytic reactor as CAPE-OPEN unit operation model applied to ethene production by ethanol dehydration process. *Chem. Eng. Trans.* 2011, 24, 403–408.
- (17) Pinho, A. R.; Cabral, J. A. R.; Leite, L. F. Ethanol: A green raw material for the petrochemical industry. 234th ACS National Meeting; Boston, MA, 2007.
- (18) Andriago, P.; Bagatin, R.; Pagani, G. Fixed bed reactors. *Catal. Today* 1999, 52 (2–3), 197–221.
- (19) Henning, G. P.; Perez, G. A. Parametric sensitivity in fixed-bed catalytic reactors. *Chem. Eng. Sci.* 1986, 41 (1), 83–88.
- (20) Froment, G. F. Fixed bed catalytic reactors-current design status. *Ind. Eng. Chem.* 1967, 59 (2), 18–27.
- (21) Lopez, A. S.; De Lasa, H. I.; Porras, J. A. Parametric sensitivity of a fixed bed catalytic reactor: Cooling fluid flow influence. *Chem. Eng. Sci.* 1981, 36 (2), 285–291.
- (22) Froment, G. F. Fixed bed catalytic reactors. *Chem. Ing. Technik* 1974, 46 (9), 374–386.
- (23) Froment, G. F.; Bischoff, K. B.; De Wilde, J. *Chemical Reactor Analysis and Design*, 3rd ed.; John Wiley & Sons Inc.: Hoboken, 2010.
- (24) Barrocas, H. V. V.; De Assis, R. C.; Da Silva, J. B. C. M. Process for preparing ethene. US Patent 4232179, 1980-11-04.
- (25) Tavoulareas, E. S. Fluidized-bed combustion technology. *Annu. Rev. Energy Environ.* 1991, 16, 25–57.
- (26) Zhang, W. A review of techniques for the process intensification of fluidized bed reactors. *Chin. J. Chem. Eng.* 2009, 17 (4), 688–702.
- (27) Tsao, U.; Zasloff, H. B. Production of ethylene from ethanol. US Patent 4134926, 1979-01-16.
- (28) Zhao, S. F. Simulation and optimizing of heat exchange networks for fuel ethanol separation technology, Tianjin University, Tianjin, China, 2006.
- (29) Wang, J.; Zhang, S. W. Investigation on energy saving in the process of ethylene from fermented alcohol. *Energy Conserv. Technol.* 2008, 26 (148), 148–150–167.
- (30) Kagyrmanova, A. P.; Chumachenko, V. A.; Korotkikh, V. N.; Kashkin, V. N.; Noskov, A. S. Catalytic dehydration of bioethanol to ethylene: Pilot-scale studies and process simulation. *Chem. Eng. J.* 2011, 176–177, 188–194.
- (31) Shi, B. C.; Davis, B. H. Alcohol dehydration: Mechanism of ether formation using an alumina catalyst. *J. Catal.* 1995, 157 (2), 359–367.
- (32) Knözinger, H.; Bühl, H.; Kochloef, K. The dehydration of alcohols on alumina: XIV. Reactivity and mechanism. *J. Catal.* 1972, 24 (1), 57–68.
- (33) Henne, A. L.; Matuszak, A. H. The dehydration of secondary and tertiary alcohols. *J. Am. Chem. Soc.* 1944, 66 (10), 1649–1652.
- (34) Takezawa, N.; Hanamaki, C.; Kobayashi, H. The mechanism of dehydrogenation of ethanol on magnesium oxide. *J. Catal.* 1975, 38 (1–3), 101–109.
- (35) Chiang, H.; Bhan, A. Catalytic consequences of hydroxyl group location on the rate and mechanism of parallel dehydration reactions of ethanol over acidic zeolites. *J. Catal.* 2010, 271 (2), 251–261.
- (36) Costa, E.; Uguina, A.; Aguado, J.; Hernandez, P. J. Ethanol to gasoline process: Effect of variables, mechanism, and kinetics. *Ind. Eng. Chem. Process Des. Dev.* 1985, 24 (2), 239–244.
- (37) Saito, Y.; Niiyama, H. Reaction mechanism of ethanol dehydration on/in heteropoly compounds: Analysis of transient behavior based on pseudo-liquid catalysis model. *J. Catal.* 1987, 106 (2), 329–336.
- (38) Okuhara, T.; Arai, T.; Ichiki, T. Dehydration mechanism of ethanol in the pseudoliquid phase of $H_{3-x}Cs_xPW_{12}O_{40}$. *J. Mol. Catal.* 1989, 55 (1), 293–301.
- (39) Dömöke, M.; Tótha, M.; Raskó, J.; Erdőhelyi, A. Adsorption and reactions of ethanol and ethanol–water mixture on alumina-supported Pt catalysts. *Appl. Catal., B* 2007, 69 (3–4), 262–272.
- (40) Rajakumar, B.; Reddy, K. P. J.; Arunan, E. Thermal decomposition of 2-fluoroethanol: single pulse tube and ab initio studies. *J. Phys. Chem. A* 2003, 107 (46), 9782–9793.
- (41) Galvita, V. V.; Semin, G. L.; Belyaev, V. D.; Semikolenova, V. A.; Tsiakaras, P.; Sobyannin, V. A. Synthesis gas production by steam reforming of ethanol. *Appl. Catal., A* 2001, 220 (1–2), 123–127.
- (42) Brey, W. S.; Krieger, K. A. The surface area and catalytic activity of aluminum oxide. *J. Am. Chem. Soc.* 1949, 71 (11), 3637–3641.

- (43) Tanabe, K.; Misono, M.; Ono, Y.; Hattori, H. *New Solid Acids and Bases: Their Catalytic Properties*; Kodansha Ltd., Elsevier Science Publishers B.V.: Amsterdam, Tokyo, 1989.
- (44) Smith, M. B.; March, J. *March's Advanced Organic Chemistry: Reactions, Mechanisms, and Structure*, 6th ed.; Wiley-Interscience A John Wiley & Sons, Inc.: USA, 2006.
- (45) Noller, H.; Thomke, K. Transition states of catalytic dehydration and dehydrogenation of alcohols. *J. Mol. Catal.* **1979**, *6* (5), 375–392.
- (46) Ramesh, K.; Hui, L. M.; Han, Y. F. Structure and reactivity of phosphorous modified H-ZSM-5 catalysts for ethanol dehydration. *Catal. Commun.* **2009**, *10* (5), 567–571.
- (47) Zhang, J.; Zhou, D. H.; Ni, D. The theoretical calculation study on reaction mechanism of ethylene dimerization on H-ZSM-5 zeolite catalyst. *Chin. J. Catal.* **2008**, *2* (8), 715–719.
- (48) Helmut, K.; Rainer, K. The dehydration of alcohols over alumina: I. The reaction scheme. *J. Catal.* **1966**, *5* (2), 264–270.
- (49) Knözinger, H. Dehydration of Alcohols on Aluminum Oxide. *Angew. Chem., Int. Ed. Engl.* **1968**, *7* (10), 791–805.
- (50) Arai, H.; Take, J. I.; Saito, Y.; Yoneda, Y. *J. Catal.* **1967**, *9* (2), 146–153.
- (51) Jain, J. R.; Pillai, C. N. Catalytic dehydration of alcohols over alumina: Mechanism of ether formation. *J. Catal.* **1967**, *9* (4), 322–330.
- (52) Abd El-Salaam, K. M.; Hassan, E. A. Active surface centres in a heterogeneous CdO catalyst for ethanol decomposition. *Surf. Technol.* **1982**, *16* (2), 121–128.
- (53) Gao, H. B. *Organic Chemistry*, 4th ed.; Higher Education Press: Beijing, 2005.
- (54) Hu, T. G.; Cheng, K. K.; Zhang, J. A.; Wang, Y. G. Research progress of catalytic dehydration of bio-ethanol into ethylene. *Mod. Chem. Ind.* **2007**, *27* (2), 96–99.
- (55) Pan, L. R. Development review of catalysts for ethanol dehydration to produce ethylene. *Speciality Petrochem.* **1986**, *4*, 41–64.
- (56) Pearson, D. E.; Tanner, R. D.; Picciotto, I. D.; Sawyer, J. S.; Cleveland, J. H., Jr. Phosphoric acid systems. 2. Catalytic conversion of fermentation ethanol to ethylene. *Prod. Res. Dev.* **1981**, *20* (4), 734–740.
- (57) Adkins, H.; Perkins, P. P. Dehydration of alcohols over alumina. *J. Am. Chem. Soc.* **1925**, *47* (4), 1163–1167.
- (58) Pines, H.; Haag, W. O. Alumina: catalyst and support. I. Alumina, its intrinsic acidity and catalytic activity. *J. Am. Chem. Soc.* **1960**, *82* (10), 2471–2483.
- (59) Zhu, X. R. Dehydration of bioethanol into ethylene over modified nano-scale HZSM-5 catalysts. Dalian University of Technology: Dalian, 2007.
- (60) Gao, Z.; He, M. Y.; Dai, Y. Y. *Zeolite Catalysis and Separation Technology*; China Petrochemical Press: Beijing, 1999.
- (61) Roca, F. F.; De Mourgues, L.; Trambouze, Y. Catalytic dehydration of ethanol over silica-alumina. *J. Catal.* **1969**, *14* (2), 107–113.
- (62) Xu, J. Development of high selectivity catalysts for preparing ethylene from ethanol dehydration. *Chem. React. Eng. Technol.* **2011**, *27* (1), 64–67.
- (63) Kochar, N. K.; Merims, R.; Padia, A. S. Ethylene from ethanol. *Chem. Eng. Prog.* **1981**, *77* (6), 66–70.
- (64) Ezzo, E. M.; Elnabarawy, T.; Youssef, A. M. Studies on the mixed oxide catalyst $\text{Al}_2\text{O}_3\text{--Cr}_2\text{O}_3$ I. Surface properties. *Surf. Technol.* **1979**, *9* (2), 111–118.
- (65) Ezzo, E. M.; El-Shobaky, G. A.; Selim, M. M. Catalytic conversion of alcohols on $\text{Al}_2\text{O}_3\text{--Cr}_2\text{O}_3$ catalysts I: The catalytic decomposition of ethanol. *Surf. Technol.* **1980**, *10* (1), 47–54.
- (66) Ezzo, E. M.; Yousef, E. M.; Mazhar, H. S. Studies of the mixed oxide catalyst $\text{Al}_2\text{O}_3\text{--Cr}_2\text{O}_3$ II: The catalytic conversion of ethanol. *Surf. Technol.* **1981**, *14* (1), 65–72.
- (67) Kojima, M.; Takahiro, A.; Yukio, A. Catalyst for production of ethylene from ethanol. US Patent. 4302357, 1981-11-24.
- (68) El-Katatny, E. A.; Halawy, S. A.; Mohamed, M. A. Mohamed. Recovery of ethene-selective $\text{FeOx}/\text{Al}_2\text{O}_3$ ethanol dehydration catalyst from industrial chemical wastes. *Appl. Catal. A: Gen.* **2000**, *199* (1), 83–92.
- (69) Chen, G. W.; Li, S. L.; Jiao, F. J. Catalytic dehydration of bioethanol to ethylene over $\text{TiO}_2/\gamma\text{-Al}_2\text{O}_3$ catalysts in microchannel reactors. *Catal. Today* **2007**, *125* (1–2), 111–119.
- (70) Yu, Y. X.; Liu, B. W. Research and production of NC1301 type catalyst for ethanol dehydration to produce ethylene. *J. Chem. Ind. Eng.* **1995**, *16* (2), 8–10.
- (71) Li, Y.; Chen, X. C.; Sun, W.; Liu, S. W.; Hou, W. Experimental study of the catalytic dehydration of ethanol to ethylene on a $\gamma\text{-Al}_2\text{O}_3$ catalyst. *J. Beijing Univ. Chem. Technol.* **2007**, *34* (5), 449–452.
- (72) Xu, R. R.; Pang, W. Q.; Yu, J. H. *Zeolite and Porous Materials Chemistry*; Science Press: Beijing, 2004.
- (73) Li, Y. N.; Jin, Z. S.; Yang, W. M. Research status of dehydration of ethanol into ethylene over zeolite. *Chem. Ind. Eng. Progress* **2009**, *28* (1), 67–72.
- (74) Argauer, R. J.; Landolt, G. R. Crystalline zeolite ZSM-5 and method of preparing the same. US Patent 3702886, 1972-11-14.
- (75) Dwyer, F. G.; Jenkins, E. E. Crystalline silicates and method of preparing the same. US Patent 3941871, 1976-03-02.
- (76) Olson, D. H.; Kokotailo, G. T.; Lawton, S. L.; Meier, W. M. Crystal structure and structure-related properties of ZSM-5. *J. Phys. Chem.* **1981**, *85* (15), 2238–2243.
- (77) Koningsveld, H. V.; Jansen, J. C.; Bekkum, H. V. The monoclinic framework structure of zeolite H-ZSM-5. Comparison with the orthorhombic framework of as-synthesized ZSM-5. *Zeolites* **1990**, *10* (4), 235–242.
- (78) Hao, T. Dehydration of dilute aqueous solution of ethanol into ethylene over ZSM-5 zeolite catalysts. *Petrochem. Technol.* **1985**, *14* (2), 92–93.
- (79) Le Van Mao, R.; Nguyen, T. M.; McLaughlin, G. P. The Bioethanol-to-ethylene (B.E.T.E.) process. *Appl. Catal.* **1989**, *48* (2), 265–277.
- (80) Phillips, C. B.; Datta, R. Production of ethylene from hydrous ethanol on H-ZSM-5 under mild conditions. *Ind. Eng. Chem. Res.* **1997**, *36* (11), 4466–4475.
- (81) Pan, L. R.; Li, H. X. Research and production of new catalyst for ethanol dehydration to produce ethylene (I). *Petrochem. Technol.* **1985**, *14* (3), 154–157.
- (82) Pan, L. R.; Li, H. X. NKC-03A catalyst for ethanol dehydration to ethylene. *Petrochem. Technol.* **1987**, *16* (11), 764–768.
- (83) Le Van Mao, R.; Dao, L. H. US patent 4698452, 1987-10-06.
- (84) Bun, S.; Nishiyama, S.; Tsuruya, S.; Masai, M. Ethanol conversion over ion-exchanged ZSM-5 zeolites. *Appl. Catal.* **1990**, *59* (1), 13–29.
- (85) Takahara, I.; Saito, M.; Inaba, M.; Murata, K. Dehydration of ethanol into ethylene over solid acid catalysts. *Catal. Lett.* **2005**, *105* (3–4), 249–252.
- (86) Jia, O. Y.; Kong, F. X.; Su, G. D.; Hu, Y. C.; Song, Q. L. *Catal. Lett.* **2009**, *132* (1–2), 64–74.
- (87) Ramesh, K.; Jie, C.; Han, Y. F.; Borgna, A. Synthesis, characterization, and catalytic activity of phosphorus modified H-ZSM-5 catalysts in selective ethanol dehydration. *Ind. Eng. Chem. Res.* **2010**, *49* (9), 4080–4090.
- (88) Lok, B. M.; Messina, C. A.; Patton, R. L. Silicoaluminophosphate molecular sieves: another new class of microporous crystalline inorganic solids. *J. Am. Chem. Soc.* **1984**, *106* (20), 6092–6093.
- (89) Djieugoue, M. A.; Prakash, A. M. Catalytic study of methanol-to-olefins conversion in four small-pore silicoaluminophosphate molecular sieves: Influence of the structural type, nickel incorporation, nickel location, and nickel concentration. *J. Phy. Chem. B.* **2000**, *104* (27), 6452–6461.
- (90) Wang, D. Y.; Li, J. L.; Fan, M. G. Synthesis of SAPO-34 molecular sieve and its application to the dehydration of ethanol. *Chin. J. Catal.* **1992**, *13* (3), 234–236.
- (91) Dahl, I. M.; Wendelbo, R.; Andersen, A.; Akporiaye, D.; Mostada, H.; Fuglerudet, T. The effect of crystallite size on the activity and selectivity of the reaction of ethanol and 2-propanol over SAPO-34. *Microporous Mesoporous Mater.* **1999**, *29* (1–2), 159–171.
- (92) Zhang, X.; Wang, R. J.; Yang, X. X.; Zhang, F. B. Comparison of four catalysts in the catalytic dehydration of ethanol to ethylene. *Microporous Mesoporous Mater.* **2008**, *116* (1–3), 210–215.

- (93) Zhou, T.; Shi, X. X.; Wu, L. N.; Cui, Q.; Wang, H. Y.; Hu, Y.; Huang, H. Catalytic Properties of SAPO-11/HZSM-5 for ethanol dehydration to ethylene. *J. Chem. Eng. Chin. Univ.* **2011**, *25* (3), 453–458.
- (94) Pan, F.; Wu, Y. L.; Zhang, J. A.; Tong, J. M. Development status of ethylene production by catalytic dehydration of fermented alcohol. *Modern Chem. Ind.* **2006**, *10* (S2), 27–31.
- (95) Mizuno, N.; Misono, M. Heteropolyacid catalysts. *Curr. Opin. Solid State Mater. Sci.* **1997**, *2* (1), 84–89.
- (96) Wang, N. J.; Li, C. H.; Gu, C. X. Recent developments in heteropoly acid and their supported catalysts. *Ind. Catal.* **2007**, *15* (10), 1–4.
- (97) Zhao, B. L.; Zhao, B. Z. Ethylene preparation with alcohol dehydration catalized by heteropololy acid. *J. Northeast Normal Univ. (Nat. Sci. Ed.)* **1995**, *1*, 70–72.
- (98) Vázquez, P.; Pizzio, L.; Cáceres, C. Silica-supported heteropolyacids as catalysts in alcohol dehydration reactions. *J. Mol. Catal. A: Chem.* **2000**, *161* (1–2), 223–232.
- (99) Haber, J.; Pamin, K.; Matachowski, L.; Napruszewska, B.; Poltowicz, J. Potassium and silver salts of tungstophosphoric acid as catalysts in dehydration of ethanol and hydration of ethylene. *J. Catal.* **2002**, *207* (2), 296–306.
- (100) Varislia, D.; Dogua, T.; Dogu, G. Ethylene and diethyl-ether production by dehydration reaction of ethanol over different heteropolyacid catalysts. *Chem. Eng. Sci.* **2007**, *62* (18–20), 5349–5352.
- (101) Xu, X. Y.; Xie, A. J. Industrial production of ethanol dehydration to ethylene. *China Surfact. Deterg. Cosmet.* **1988**, *5*, 23–28.
- (102) Kochar, N. K. Ethylene from ethanol. *Chem. Eng.* **1980**, *28* (1), 80–81.